

Milestones for Placement Portal Application – V2

Modern Application Development II

Project: Placement Portal Application V2 (MAD-II)

The project is divided into **0th milestone, Core Requirements (8 Milestones), and Recommended Functionalities & Optional Enhancements (2 Milestones)**.

All core milestones can be completed in any order.

Recommended and optional enhancements should only be started after the core milestones are done.

Each milestone includes a Git commit message. When completed, **use a unique commit message and push your updates to GitHub.**

Milestone 0: GitHub Repository Setup (Mandatory) **Deadline: 14/07/2026**

 Expected Time: 1 day
 Completion Progress: 5%

Create a **private GitHub repository**.

Add **README.md** with a short project description.

Add **.gitignore** for Python, Flask, Vue, SQLite, Redis cache files. (not compulsory).

Commit and push initial setup to the repository.

Refer to the [git tracker document](#) to add **MADII-cs2006** as your collaborator, so that we can track your project. (**Requests will be accepted after term starts.**)

Github Adding Collaborator Video: [Adding Github Collaborator using VSCode and WSL](#)

This milestone needs to be completed by 14th July 2026.

Git Commit Message: Milestone-0 PPA-V2 Setup

Core Requirements (8 Milestones)

Milestone: Database Models and Schema Setup (Flask)

✓ Expected Time: 5–7 days
📊 Completion Progress: 15%

Create models for:

User (role: Admin / Company / Student)

Company (profile, industry, location, etc.)

Student (profile, education, skills, resume, etc.)

Job Position (company_id, title, salary, skills_required, etc.)

Application (student_id, job_id, status, date, etc.)

Placement (student_id, company_id, position, salary, joining_date, etc.)

Any other if you need.

Define relationships (1:n, n:n where required, e.g., Company–Job Position, Student–Application, Application–Placement).

Ensure Admin user is **pre-created programmatically** through a Python script/file (using SQLite query or SQLAlchemy ORM).

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Authentication and Role-Based Access (Admin/Company/Student) (Flask+Vue)



Expected Time: 5 days



Completion Progress: 10%

Implement authentication using **Flask-Security** or **JWT-based tokens**.

Students: self-register + login.

Companies: register + login only (approved by Admin).

Admin: predefined login only (no registration).

Redirect users to **role-specific dashboards** after login (Admin, Company, Student).

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Admin Dashboard and Management (Flask+Vue)

✓ Expected Time: 7–9 days

📊 Completion Progress: 15%

Admin functionalities:

Dashboard showing total students, companies, job postings, and applications.

Approve and Remove Company profiles.

Approve and Remove job posting/placement drives created by Companies.

Search companies (by name / industry).

Search students (by name / ID / contact).

View and manage all job postings and applications.

Blacklist/Deactivate companies and students from the system.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Company Dashboard and Job/Application Management (Flask+Vue)

✓ Expected Time: 7 days

📊 Completion Progress: 15%

Company functionalities:

Companies will register their profile.

Companies can only access the dashboard when approved by admin.

Dashboard showing job postings, received applications, and shortlisted candidates.

Post new job positions with required skills, experience, salary, and benefits.

View list of applicants for each job.

Shortlist or reject applicants with feedback.

Manage job posting status (Active / Closed).

Schedule interviews with shortlisted candidates.

Share acceptance or rejection status(Shortlisted / Selected / Rejected) to applicants.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Student Dashboard and Job Application System (Flask+Vue)



Expected Time: 7 days



Completion Progress: 13%

Student functionalities:

Register, log in, and update profile (education, skills, resume, experience).

View and search job postings by company, position, or required skills.

Apply for jobs and track application status.

View applied jobs with detailed application status.

View interview schedules and feedback from companies.

Download offer letters or placement confirmations.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Job Application History and Status Tracking (Flask+Vue)

 Expected Time: 5–6 days

 Completion Progress: 12%

Store and display complete application and placement history.

Prevent duplicate applications for the same job posting.

Ensure only approved companies can create placement drives.

Ensure students can view and apply only to approved placement drives.

Maintain status updates (Applied / Shortlisted / Interview / Offer / Rejected / Placed).

Admin and Company can view student profiles and applications; Students can view their own records.

Git Commit Message: **USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE**

Milestone: Backend Jobs – Interview Reminders, Placement Reports, and Triggered Jobs (CSV Export using Celery + Redis) (Flask)

 Expected Time: 7–8 days
 Completion Progress: 10%

Setup Celery workers, Celery Beat, and Redis server.

Interview Reminder Job: send reminders (Email / GChat / SMS) to students with scheduled interviews.

Placement Report Job: generate monthly reports for companies (HTML / PDF) with application statistics, placements, and analytics.

User-triggered CSV Export: students and companies can export application and placement history asynchronously; alert sent once the batch job is complete.

Git Commit Message: **USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE**

Milestone: API Performance Optimization and Caching Using Redis (Flask)

 Expected Time: 3 days
 Completion Progress: 5%

Use **Redis caching** for API optimization.

Cache frequently used endpoints (job listings, company search, student search).

Implement proper cache expiry and refresh policies.

Git Commit Message: **USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE**

Recommended / Optional Enhancements (2 Milestones)

These milestones are recommended but not mandatory.

Milestone: UI/UX Enhancements and PWA Features (Vue)

✓ Expected Time: 6 days

Improve **UI styling** using Bootstrap & VueJS.

Make the web app mobile-responsive.

Implement **frontend validation** (JavaScript) and **backend validation** (Flask).

Add "Add to Home Screen" (PWA like behavior).

Consistent design across devices.

Git Commit Message: **USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE**

Milestone: Reports, Charts and ATS (Vue)

✓ Expected Time: 10 days

Integrate Chart.js for analytics (placement trends, job demand by skills, application funnels).

Public Landing Dashboard Page (pre-login) showing **aggregated monthly placement statistics** (read-only, no sensitive data).

Create a ATS-style resume screener which can be used by both students and companies

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Final Submission

Milestone: Final Project Submission



Expected Time: 2 days

After Completion Progress: 100%

Submit **ZIP with code + 3–5 page report**.

Submit a 3-10 minute demo video (Google Drive with access to **anyone with the link**).

Include a **video link in the report**.

Include a **declaration of AI usage %** in the same pdf report (**if used or not declaration is mandatory**).

Use this video [How to Submit App Dev 1 & 2 Projects | Full Submission Guide + Viva Workflow | IITM BS](#) on how to submit the app in the portal.

Git Commit Message: **Milestone-PPA-V2 Final-Submission**

Tracking Progress

To track progress, you should:

Use Git commits for each milestone.

Test functionality after completing each milestone.

Log issues encountered and their resolutions in a **README.md** or GitHub Issues section.

Timeline Overview

Project Completion Overview

Milestone	Average Time Estimate	Split (%)	Progress (%)
GitHub Setup	1 day	5%	5%
DB Models & Schema	5-7 days	15%	20%
Auth & RBAC	5 days	10%	30%
Admin Dashboard	7-9 days	15%	45%
Company Dashboard	7 days	15%	60%
Student Dashboard	7 days	13%	73%
Placement Status	5-6 days	12%	85%

Celery Jobs	7-8 days	10%	95%
Redis Caching	3 days	5%	100%

Extra Work Overview

Milestone	Time
Responsive UI/PWA	6 days
Reports, Charts & ATS	10 days

Final Submission Overview

Milestone	Average Time Estimate
Final Project Submission Details	2 days

Note:

Time estimates are averages. Spend time, as needed, to understand and complete each milestone. Completing Milestone 0 and 8 core requirements are taken for grading in Viva1. After completing the core requirements, there are 2 additional milestones for enhancements. These enhancements are not graded but can be completed for improved functionality or to be considered for the Best Course Project nomination.

Students can complete the core milestones in any order as per their approach to doing the project. They need not follow the same sequence as they appear in this document.

Students can have any number of commits in their project repository, but for a milestone to be considered as “complete”, the student must make a commit with a unique commit message of that milestone.

It is mandatory to push the entire codebase to the GitHub before you submit the code on the viva-portal.